Clinical Trials - Knowledge Graph

What I built

01

Drug Intelligence

Evidence · Competition · Geography · Multi-hop paths

02

Disease Analytics

Landscape · Design · Phase · Enrollment · Sponsor diversity

03

Sponsor Intelligence

Portfolio · Geo reach · Pipeline · Drugs · Collaborators

04

Network & Graph

Centrality · Communities · Repurposing signals

Evidence Profile

WHAT IT DOES

Aggregates every trial ever run for a drug by phase, status, condition and geography in a single query. Replaces weeks of manual registry work.

WHO USES IT

Drug developers, BD teams assessing clinical maturity before in-licensing, and medical affairs preparing HCP evidence summaries.

Geographic Footprint

WHAT IT DOES

Maps the countries actively conducting trials for a drug, exposing regional concentration risk and untapped recruitment markets.

WHO USES IT

Clinical operations optimizing site selection; market access teams forecasting country by country launch readiness.

Competitive Landscape

WHAT IT DOES

Finds every drug that shares a condition with the target compound, ranked by trial volume, also every sponsor investing in the drug.

WHO USES IT

Portfolio strategy and white space mapping; medical affairs competitive dossiers; licensing analysts sizing the field.

Multi-hop Paths

WHAT IT DOES

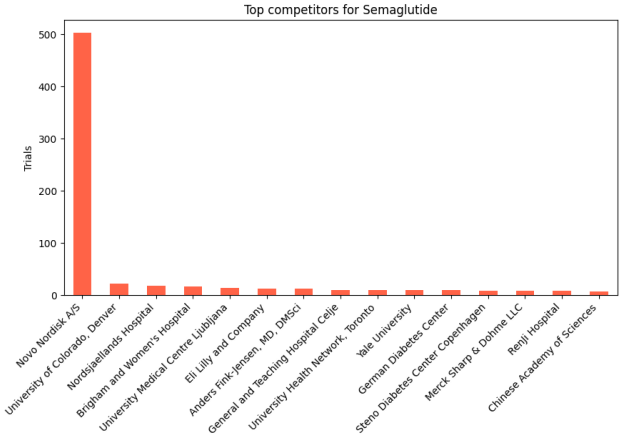
Constructs the Drug → Study → Condition → Sponsor subgraph, the structured context chunk fed to the LLM in the GraphRAG pipeline.

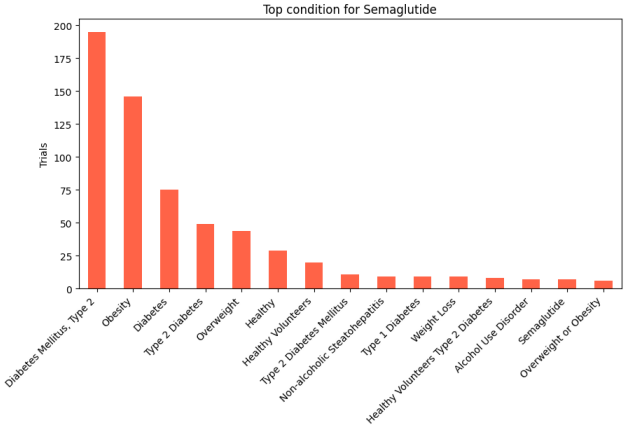
WHO USES IT

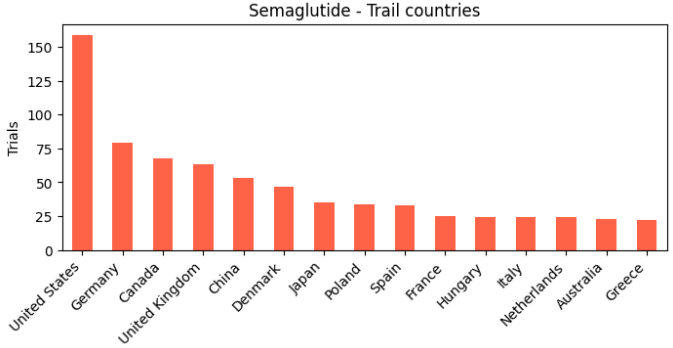
AI/ML teams building grounded clinical Q&A agents; regulatory writers who need citation ready evidence chains.

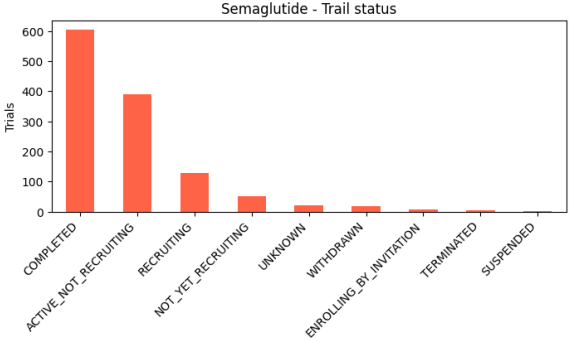
01 / DRUG INTELLIGENCE - Semaglutide

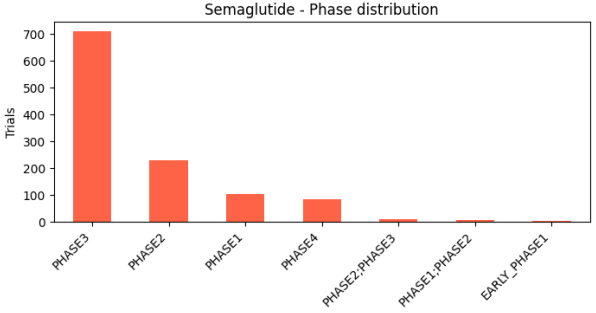
Top competitors to Semaglutide		
Metformin	Clopidogrel	Ritonavir
Zidovudine	Pioglitazone	Rosuvastatin
Sitagliptin	Aspirin	Atorvastatin
Dapagliflozin	Insulin	Itraconazole
Midazolam	Liraglutide	Lamivudine
Propofol	Insulin aspart	Prednisone
Insulin detemir		







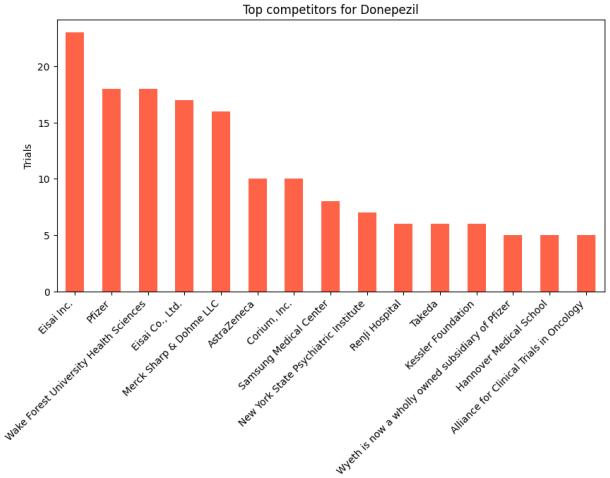


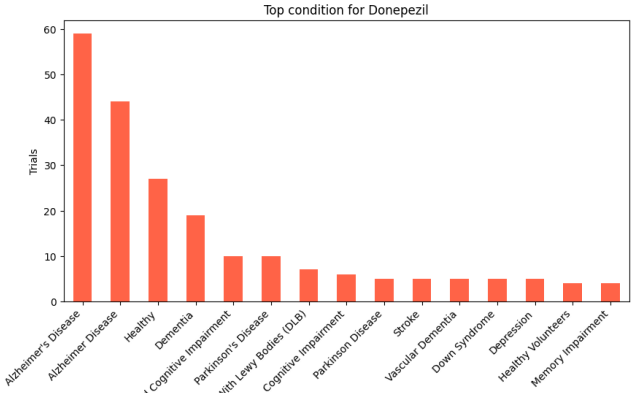


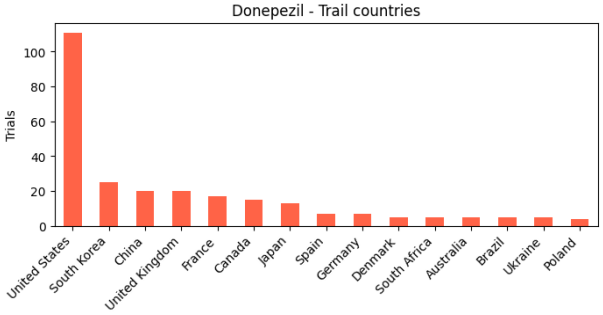
Semaglutide's trials are mostly across metabolic conditions, it is Phase 3 heavy and it has US centric geography with Novo Nordisk as main sponsor.

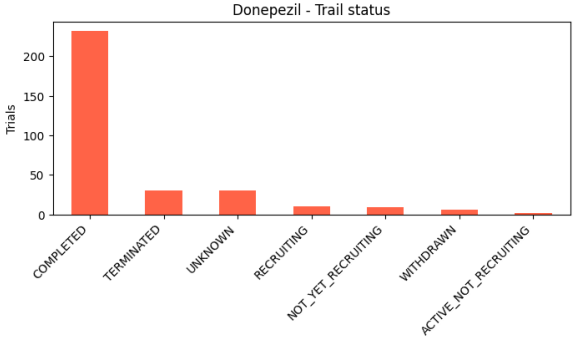
01 / DRUG INTELLIGENCE - Donepezil

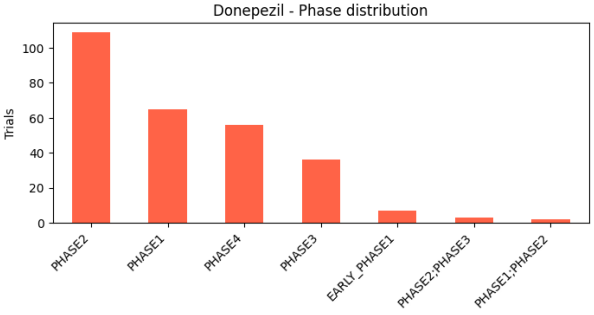
Top competitors to Donepezil		
Radiation therapy	Metformin	Paclitaxel
Carboplatin	Cyclophosphamide	Trastuzumab
Docetaxel	Methotrexate	Cisplatin
Midazolam	Aripiprazole	Olanzapine
Conventional surgery	Risperidone	Capecitabine











Donepezil's competitor list spans across oncology and antipsychotics which reveals the shared comorbidity landscape of Alzheimer's trials. The Phase 2 peak signals a disease area still seeking its breakthrough, with academic centers driving most trial activity.

1

Semaglutide dominates its disease space

Semaglutide has a lot of Phase 3 trials confirming it as the most clinically validated GLP1 agonist. The US leads site concentration with most of the trials, followed by Germany and Canada. Novo Nordisk and the University of Colorado are the dominant sponsors, with Diabetes Mellitus Type 2 and Obesity as the primary conditions.

2

Donepezil reveals unexpected competitors

Donepezil's top competitors include oncology drugs (Paclitaxel, Carboplatin) and antipsychotics (Aripiprazole, Risperidone), reflecting the shared Alzheimer's + comorbidity trial landscape. Phase 2 trials dominate, signalling a pipeline still searching for breakthrough efficacy. The US and South Korea are the leading countries.

3

Geographic concentration is a risk signal

Both drugs are heavily US centric in their trial geography, revealing recruitment concentration risk. For a drug like Semaglutide with global obesity relevance, the under representation of Asia and Africa is a meaningful gap for generalizability of trial evidence.

Treatment Landscape

WHAT IT DOES

Ranks every active drug by trial count for a condition. Shows who dominates and who is an emerging challenger.

WHO USES IT

Competitive intelligence analysts; payers building formulary evidence; investors assessing therapeutic crowding.

Enrollment Analysis

WHAT IT DOES

Histogram + phase-stratified median charts. Exposes underpowered trials and mega-trials that skew cost estimates.

WHO USES IT

Biostatisticians planning sample sizes; regulatory reviewers; health economists modelling trial cost distributions.

Trial Design Patterns

WHAT IT DOES

Breaks down allocation, blinding and arm type across all trials in a disease, shows whether the field relies on RCTs or observational evidence.

WHO USES IT

Protocol designers benchmarking against field norms; HTA bodies assessing evidence quality, meta analysts planning systematic reviews.

Sponsor Diversity

WHAT IT DOES

Classifies sponsors as Industry, Academic or Government and maps trial volumes. High academic share signals exploratory science.

WHO USES IT

Licensing managers identifying partners; grant writers mapping funding; investors gauging commercial momentum in a disease area.

Phase Progression

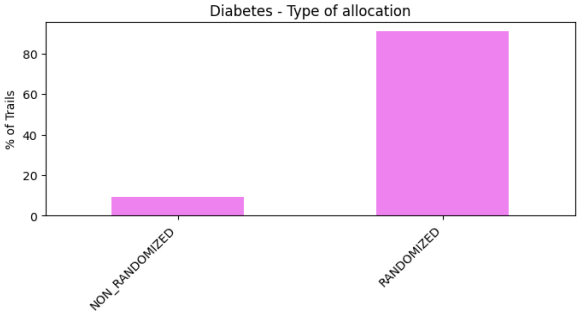
WHAT IT DOES

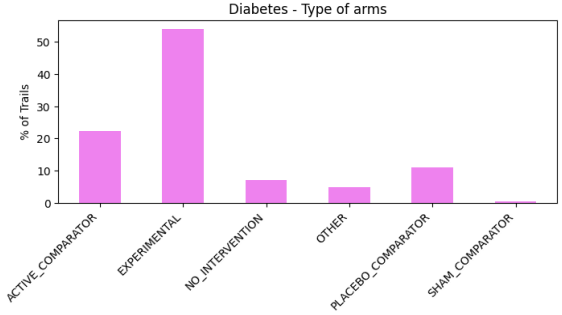
Plots trial counts by phase for the entire disease area. A healthy funnel shows I >> II >> III. Inversion signals high attrition.

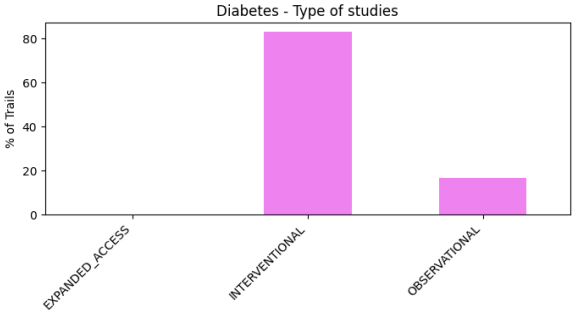
WHO USES IT

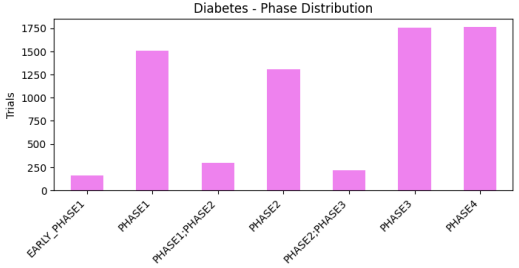
R&D strategy leads prioritizing areas; VCs screening categories; BD teams timing in-licensing to the pipeline cycle.

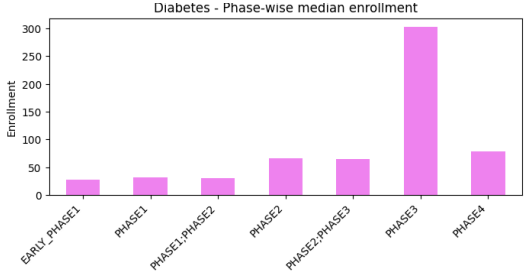
02 / DISEASE ANALYTICS - Diabetes

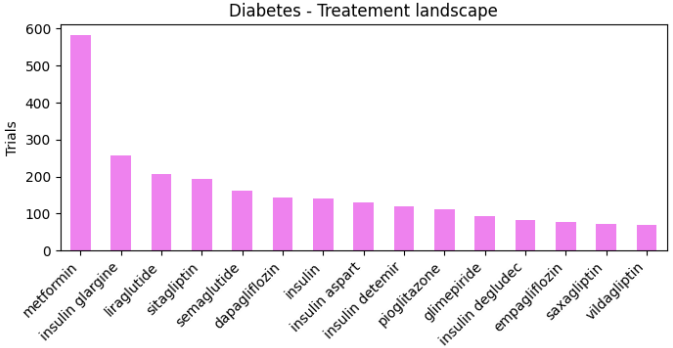


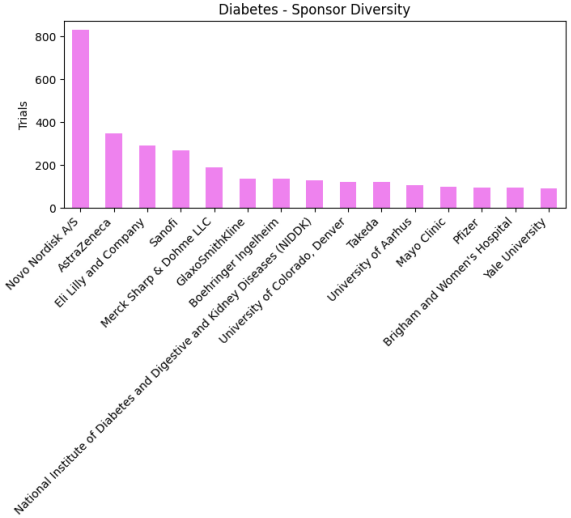






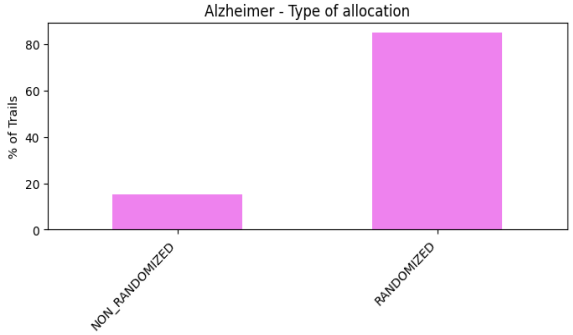


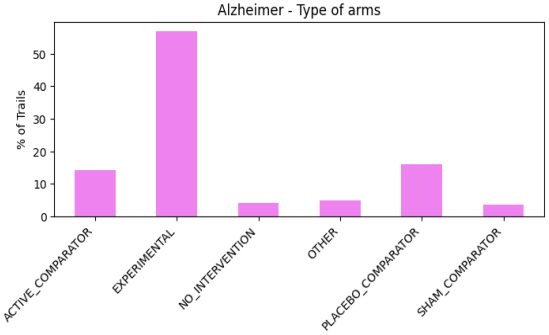


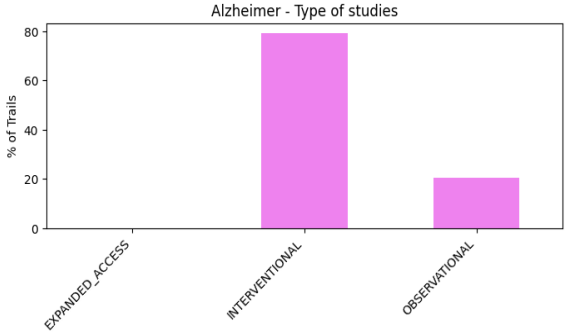


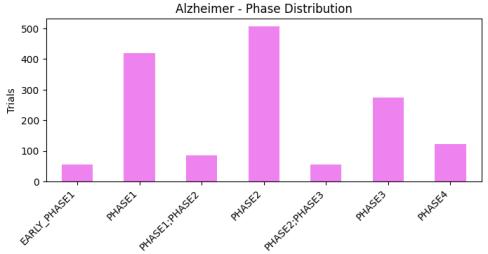
Diabetes trials represent one of the most mature, industry funded, and RCT saturated disease landscapes. The large Phase 3 enrollment medians and 85%+ randomization rate confirm this as a benchmark disease for trial design standards.

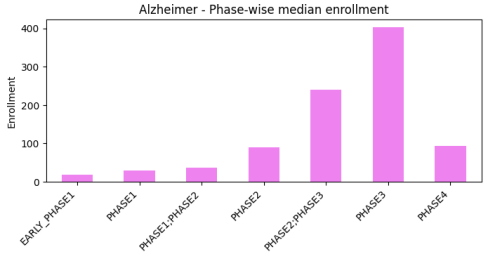
02 / DISEASE ANALYTICS - Alzheimer

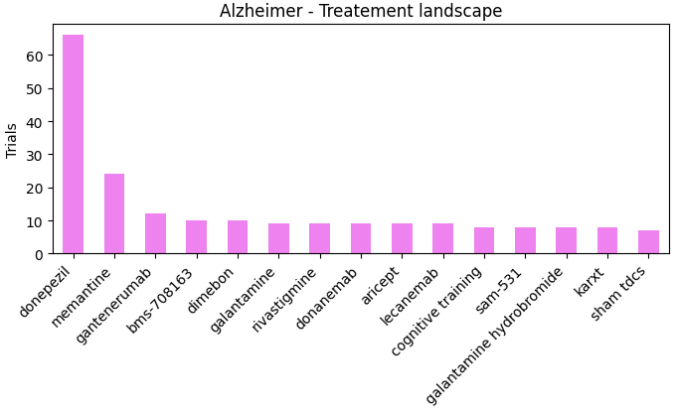


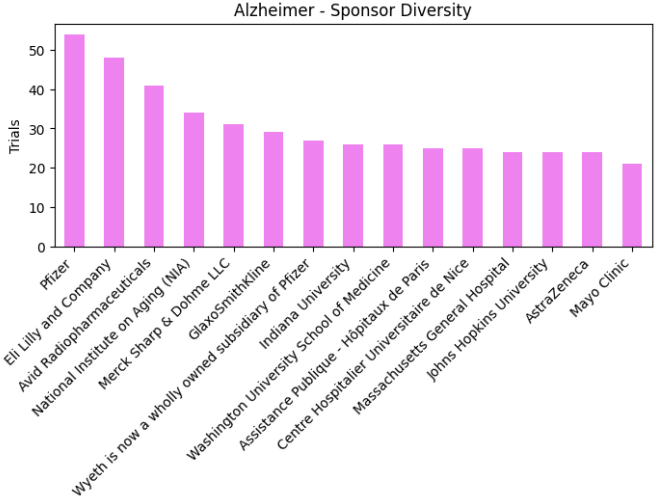












Alzheimer's trials show an inverted funnel i.e., peaking at Phase 2 with large Phase 3 enrollment requirements signaling high cost and risk. Higher academic/government sponsorship vs Diabetes reflects a disease still in the discovery phase of its clinical journey.

1

Diabetes: mature, industry led, RCT dominated

Diabetes trials show a healthy Phase 3 > Phase 2 funnel, with Phase 3 enrollment medians exceeding 300 participants. Over 85% of trials are randomized and predominantly interventional. Novo Nordisk and Eli Lilly dominate sponsor share, with Metformin and insulin glargine leading the treatment landscape.

2

Alzheimer's: early stage, fragmented, academic heavy

Alzheimer's trials peak at Phase 2, suggesting a pipeline still searching for breakout efficacy. Enrollment medians spike dramatically at Phase 3, reflecting large, expensive confirmation trials. NIH, Pfizer, and Eli Lilly co-lead, with Donepezil and Memantine as the established treatment anchors.

3

RCT evidence quality varies sharply by disease

Diabetes has ~85% randomized allocation vs Alzheimer's ~80%, but both use predominantly experimental arms. Diabetes' higher interventional share confirms stronger commercial investment and regulatory maturity. Alzheimer's higher observational share signals the field still building disease understanding.

4

Sponsor diversity signals innovation stage

Diabetes is industry dominated (Novo Nordisk, Lilly) confirming commercial viability. Alzheimer's shows higher academic/government share (NIH, academic hospitals), reflecting that the disease is still in an exploratory, grant funded phase which might be a key signal for investors and licensing teams.

Condition Portfolio

WHAT IT DOES

Ranks every disease area a sponsor runs trials in. Reveals true therapeutic focus.

WHO USES IT

BD teams mapping a counterpart's priorities before a deal; investors stress testing pipeline concentration risk.

Drug Inventory

WHAT IT DOES

Lists every distinct intervention a sponsor trials, ranked by volume. Exposes flagship compounds vs. one offs.

WHO USES IT

Portfolio strategists prioritizing investment; CMC teams forecasting manufacturing complexity; regulatory leads anticipating submission workload.

Geographic Reach

WHAT IT DOES

Maps countries where a sponsor conducts trials. Reveals regional over-reliance, regulatory arbitrage strategies, and site network maturity.

WHO USES IT

Clinical operations choosing sites; market access teams predicting first-launch markets; regulators assessing data generalizability

Collaborator Network

WHAT IT DOES

Identifies other sponsors appearing in the same trials, exposing co-development deals, consortia and academic partnerships not always announced.

WHO USES IT

BD analysts mapping partnership ecosystems; due diligence teams verifying collaborations; academic institutions identifying industry partners.

Pipeline Mix

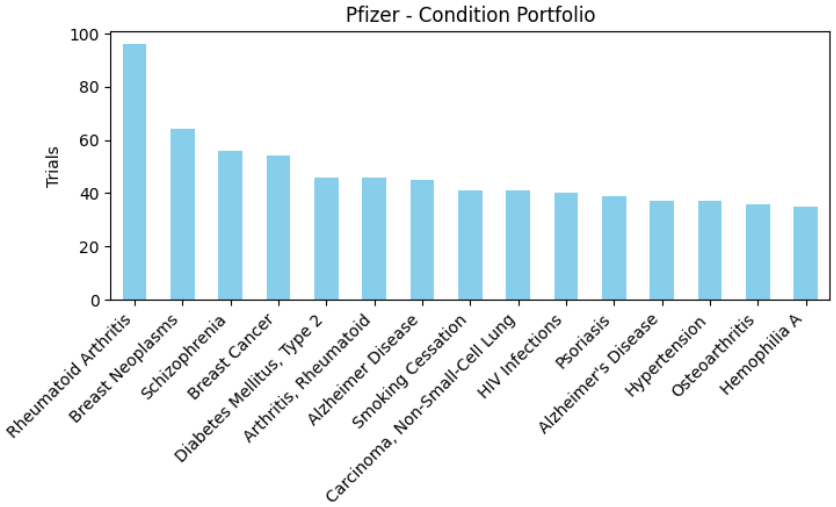
WHAT IT DOES

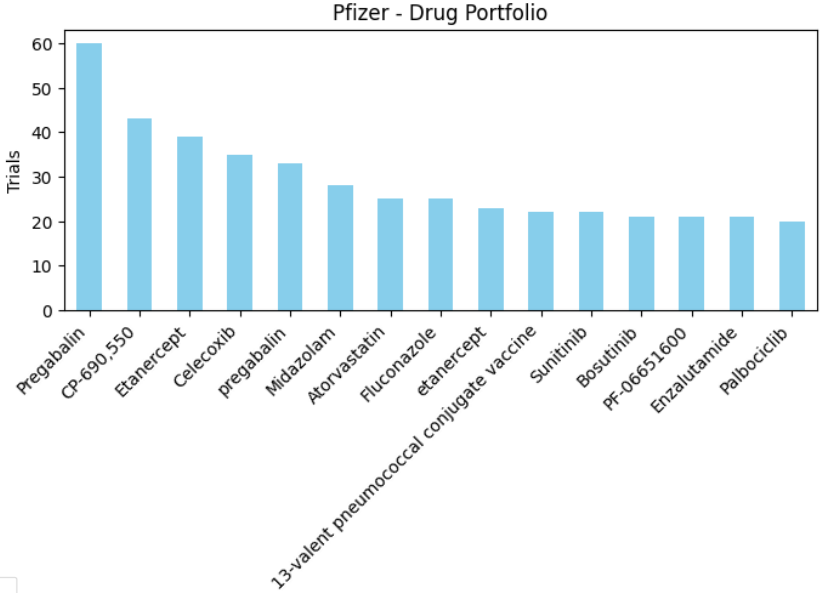
Cross tabulates trial phase against status in a stacked bar. A mature sponsor shows many COMPLETED Phase III; an early-stage one shows RECRUITING Phase I.

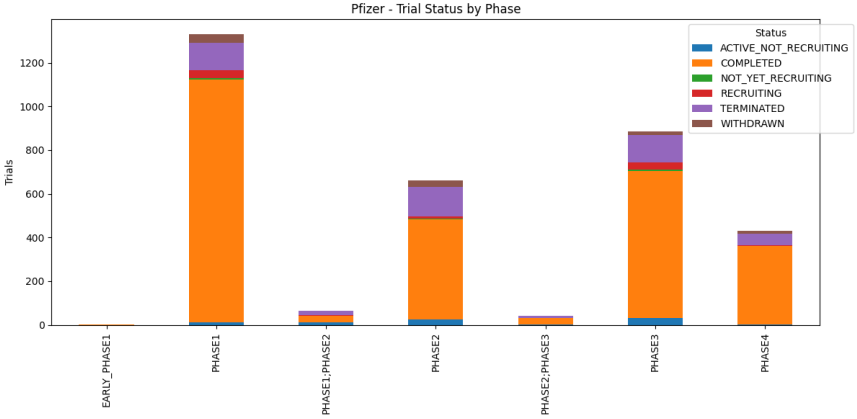
WHO USES IT

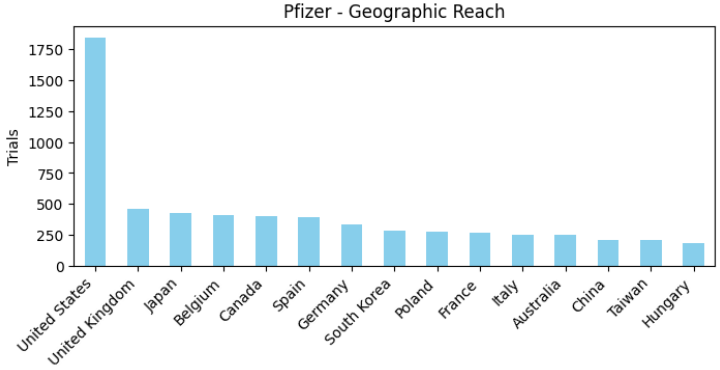
Analysts building competitive pipeline maps; investors calculating probability of success.

03 / SPONSOR INTELLIGENCE - Pfizer

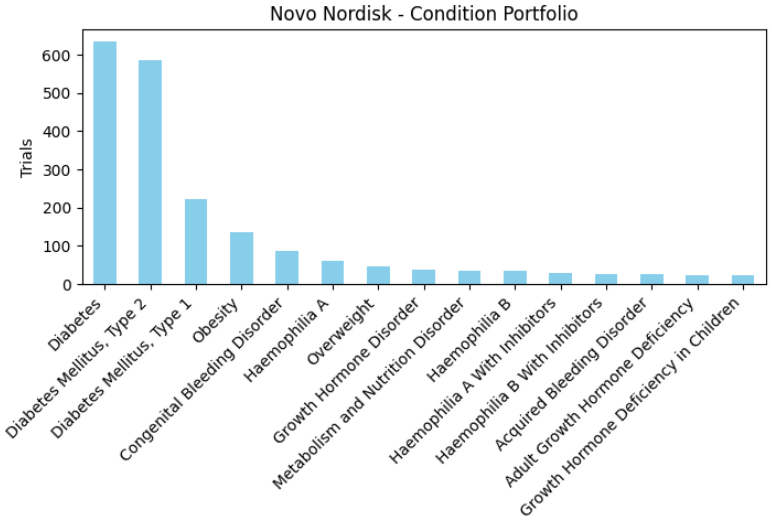


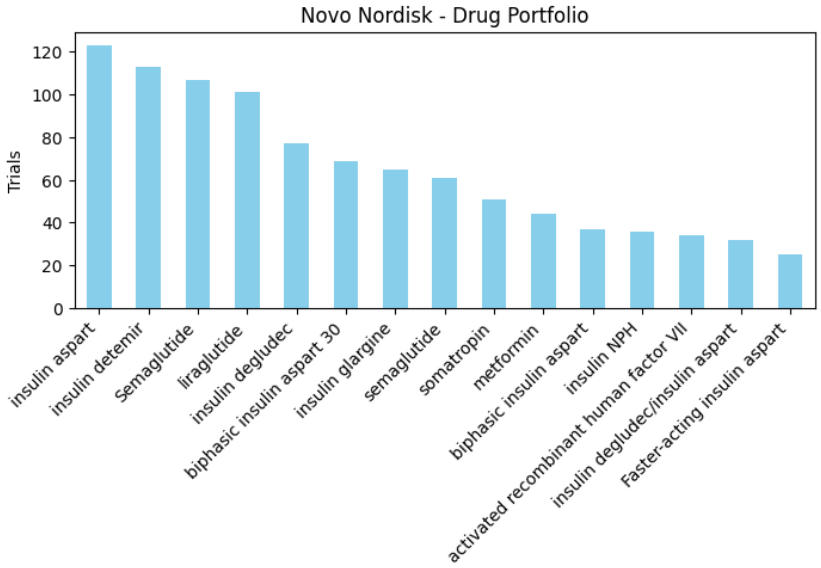


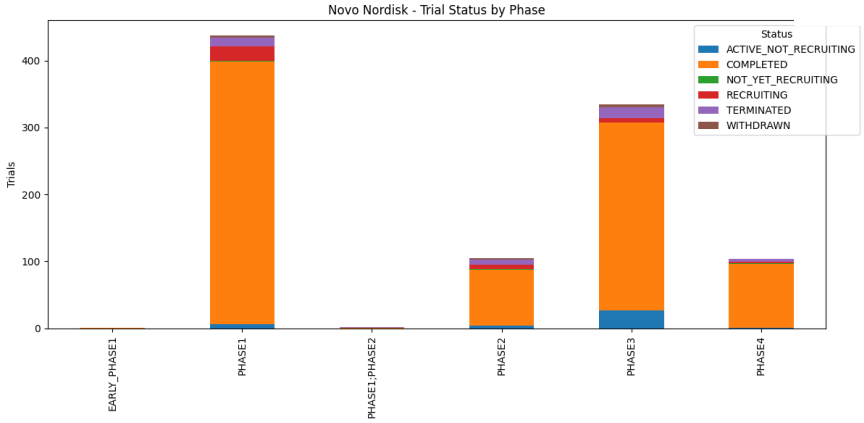


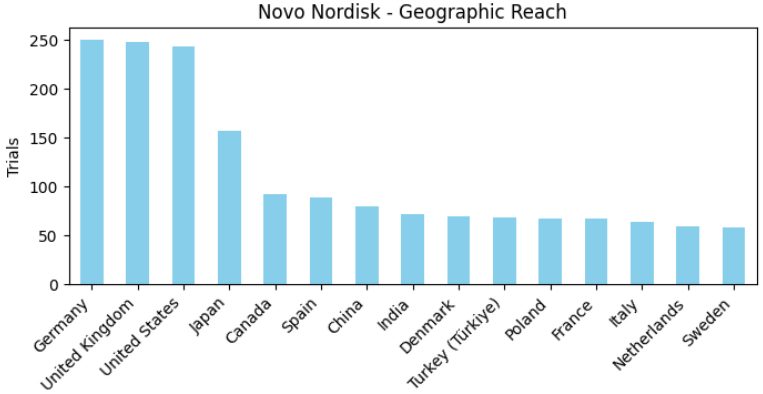


03 / SPONSOR INTELLIGENCE - Novo Nordisk









1

Pfizer: broad portfolio, late stage mature

Pfizer trials span 15+ therapeutic areas with Rheumatoid Arthritis, Breast Cancer, and Schizophrenia at the top. Its Phase 2 and Phase 3 completed bars dominate the pipeline mix, confirming a sponsor with proven commercial delivery. US and UK lead geographic reach.

2

Novo Nordisk: focused on metabolic disease

Novo Nordisk's condition portfolio is dominated by Diabetes and Obesity. Semaglutide and insulin analogues lead drug inventory. Germany tops geographic reach, reflecting European HQ. Active Phase 3 recruiting signals the next wave beyond GLP1 dominance.

Centrality Scoring

WHAT IT DOES

Computes degree centrality (unique connections) and betweenness centrality (bridge importance) across the drug condition graph. High degree nodes are therapeutically broad; high betweenness nodes are structural bridges.

WHO USES IT

Pipeline prioritization teams; drug repositioning researchers finding anchor conditions with maximum cross indication reach.

Community Detection

WHAT IT DOES

Applies the Louvain algorithm to partition the graph into dense clusters where drugs and diseases are tightly co-studied. Each community is a self organised therapeutic area.

WHO USES IT

GraphRAG pipeline builders feeding disease cluster summaries to an LLM

Drug Repurposing Signals

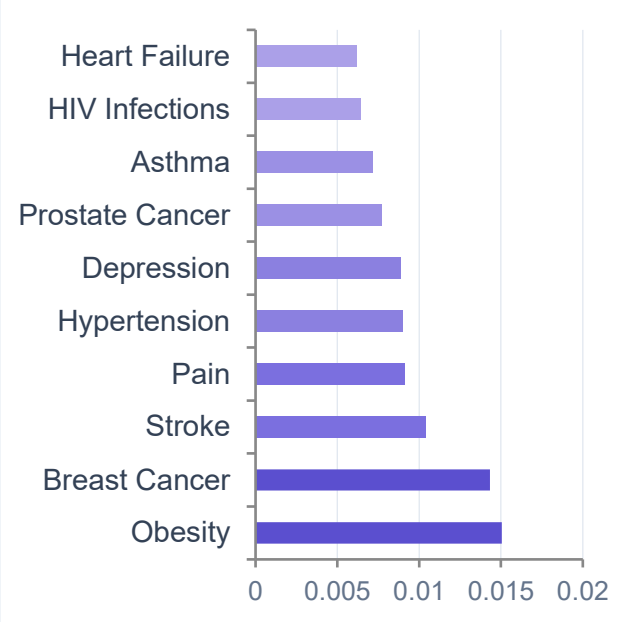
WHAT IT DOES

Identifies drugs studied across the greatest number of distinct conditions relative to trial volume. High condition breadth at moderate trial count = a drug explored beyond its label, the core repurposing signal.

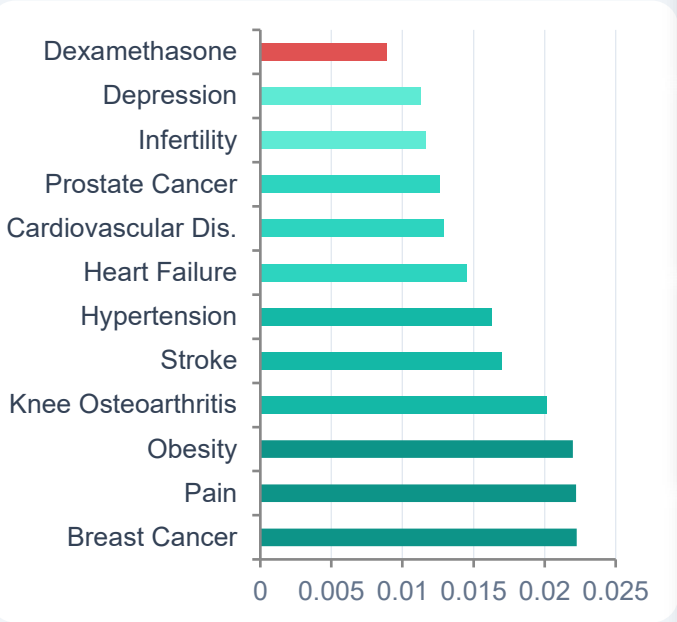
WHO USES IT

Drug repositioning researchers; biotech startups seeking off patent compounds; AI drug discovery teams building training datasets; rare disease researchers needing established safety profiles.

Degree centrality



Betweenness centrality



All top nodes are DISEASE CONDITIONS — confirming the graph's condition centric structure driven by trial volume per disease area

Obesity

Highest degree: GLP-1 agonists, SGLT2 inhibitors, bariatric, dietary, and behavioral trials all converge on this node. Reflects the global obesity epidemic driving massive trial investment.

Breast Cancer

Tested with the widest variety of drugs/chemotherapy backbones, targeted therapies (CDK4/6, HER2), immunotherapy, and hormonal agents all connect here.

Pain & Heart Failure

Despite appearing in middle of the list in degree, both score very high in betweenness, they act as bridges between analgesic, surgical, and cardiovascular clusters.

Dexamethasone (Only Drug)

The sole pharmaceutical agent in the betweenness top 25. Used in oncology premedication, COVID19 (RECOVERY trial), surgical anaesthesia, sepsis, and haematology, spanning ALL major disease communities simultaneously.

15,585	Total Communities

59.3%	Singleton or Pair Clusters

28	Mega clusters (>1,000 nodes)

The extreme skew (59% singletons) is expected in large bipartite trial graphs as rare drug condition pairs from small trials form isolated nodes, while the core knowledge concentrates in ~28 bigger communities representing major therapeutic areas.

Community #9

Metabolic & Behavioral Health

Largest cluster. Centers on Type 2 Diabetes + Depression + Obesity co-comorbidity. Dominated by behavioral interventions (CBT, motivational interviewing, mindfulness), antidepressants (Escitalopram), and lifestyle programs.

T2D · Depression · CBT · Gastric Bypass · Sleep

Community #68

Perioperative & Anaesthesia Care

Surgical drug trials: laparoscopic/bariatric surgery, cardiac and orthopaedic procedures. Key agents: Dexmedetomidine, Remifentanil, Lidocaine, Midazolam. Glucose monitoring devices link back to metabolic domain.

Dexmedetomidine · Remifentanil · Bariatric Surgery · PEEP

Community #6

Oncology + Hepatic/Metabolic Overlap

HCC and colorectal cancer trials co-occurring with metabolic agents (Metformin, Glipizide) and PD-1 inhibitors (Nivolumab, Sintilimab). Reflects trials optimizing metabolic health during cancer treatment.

Liver Cancer · Nivolumab · Metformin · PegIFN · Tamoxifen

Community #15

Neurological & Musculoskeletal Rehab

Neurodegenerative and musculoskeletal conditions with rehabilitation focus. MRI based diagnostics, exercise programs, stem cell therapies, hyperbaric oxygen. Parkinson's, cognitive decline, diabetic neuropathy.

Parkinson · Cognitive Decline · Hyperbaric O_2 · fMRI · Stem Cells

The Louvain algorithm independently recovered clinically meaningful therapeutic clusters without any expert labeling. Community boundaries align precisely with real biomedical domains, validating the KG.

Community #82

Diabetes Pharmacotherapy

Tightly focused T2D drug trials. Contains every major class: SGLT2i (Dapagliflozin, Empagliflozin), GLP-1 (Dulaglutide), DPP-4i (Sitagliptin), sulfonylureas, and insulin variants plus NASH-related interventions.

Dapagliflozin · Empagliflozin · Sitagliptin · Insulin · NASH

Community #13

Cardiovascular Disease & Prevention

Coronary artery disease, atrial fibrillation, heart failure. Atorvastatin, Aspirin (multiple doses), Evolocumab, Valsartan, Warfarin. Coronary stent comparisons (TAXUS vs Xience). SGLT2i (Dapagliflozin) entering cardiology.

Atorvastatin · Aspirin · Evolocumab · Cardiac MRI · Dapagliflozin

Community #0

Hepatitis B Virology

Almost entirely devoted to Chronic Hepatitis B (CHB). All antiviral classes: TAF/TDF, Entecavir, Lamivudine, PegIFN. Therapeutic vaccines and novel investigational agents (VIR-2218, Bepirovirsen (siRNA based)). Reflects the global HBV cure effort.

TAF/TDF · Entecavir · PegIFN · HBV Vaccine · Bepirovirsen

Community #10

Immunology, Autoimmune & Gout

Rheumatoid arthritis, gout, GvHD, and autoimmune conditions. Abatacept, Hydroxychloroquine, Rituximab, Anakinra, Allopurinol, Rasburicase, antifungals. Mix of immunosuppressants and natural anti-inflammatory compounds.

RA · Gout · Abatacept · Hydroxychloroquine · Rituximab · GvHD

Community #41

Sleep Apnea, Obesity & Pulmonary

Obstructive sleep apnea (active vs sham CPAP), polysomnography endpoints, obesity interventions, asthma. Pirfenidone (IPF), Bosentan (PAH), Roflumilast (COPD). Hyperpolarized Xe-129 MRI, cutting edge pulmonary imaging.

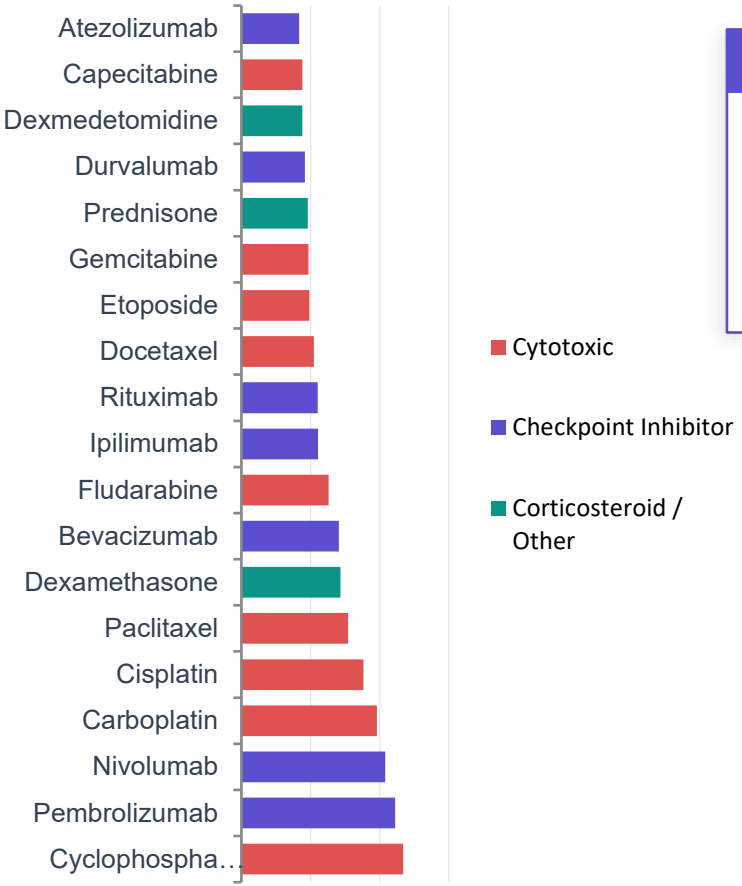
CPAP · Sleep Apnea · Pirfenidone · Bosentan · Xe-129 MRI

Community #3

Infectious Disease & Respiratory

Respiratory infections, COVID19, sepsis, influenza, critical care. Antibiotics (Vancomycin, Clindamycin), Oseltamivir, mechanical ventilation protocols, FMT, high flow nasal cannula. Nitazoxanide repurposing for COVID19.

COVID19 · Sepsis · Vancomycin · FMT · Mechanical Ventilation



Checkpoint Inhibitors

Pembrolizumab, Nivolumab, Ipilimumab, Durvalumab, Atezolizumab, expanded from melanoma/NSCLC into dozens of tumor types. Nivolumab appears in HBV and COVID19 trials, signaling cross disease immune axis potential via PD1 blockade beyond oncology.

Cytotoxic Drugs

Cyclophosphamide, Carboplatin, Cisplatin, Paclitaxel, Gemcitabine, appear in combination regimens across virtually all solid and haematological cancers. High condition breadth is partly combinatorial (many tumor subtypes tested) but also reflects genuine mechanistic versatility via DNA damage and immune modulation.

Corticosteroid

Dexamethasone, the sole non oncology drug in the top tier. Spans inflammation, surgery, sepsis, respiratory distress (ARDS), haematology. Its inclusion in the COVID19 RECOVERY trial is an example of repurposing success.

1 Disease burden drives graph structure

Degree centrality is dominated by conditions , reflecting that disease prevalence and funding priority determine how many drugs get tested against a given condition. Obesity, Breast Cancer, Stroke, and Pain are the most contested disease nodes in the entire trial network.

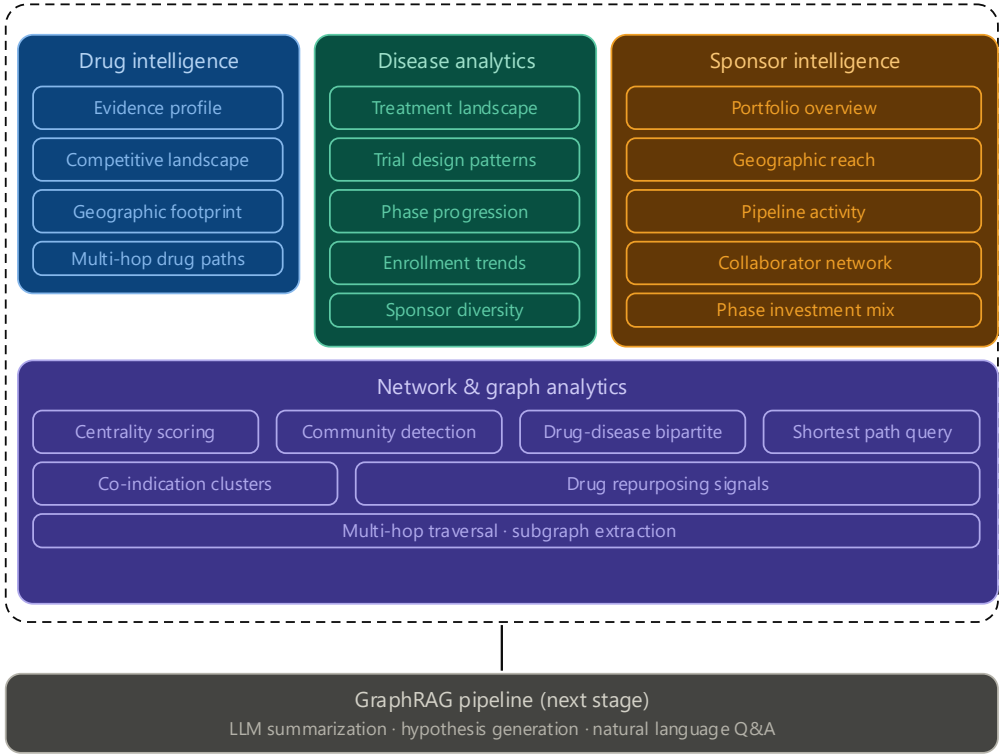
2 Betweenness reveals hidden cross domain connectors

Knee Osteoarthritis, Infertility, and Epilepsy act as structural bridges despite moderate trial volume. Dexamethasone is the only drug with this bridging role, its cross indication presence in every major disease domain makes it irreplaceable to the network's connectivity.

3 Louvain communities map to real biomedical domains

All 10 largest communities correspond to coherent therapeutic areas: metabolic disease, oncology, cardiovascular, infectious disease, and virology. The HBV community (#0) is the most specific community almost exclusively devoted to one pathogen's eradication.

4 Repurposing signals: three distinct archetypes

Cytotoxic backbone drugs (broad oncology use), checkpoint inhibitors (tumor agnostic expansion), and corticosteroid. The KG correctly surfaces Nivolumab's HBV and COVID19 appearances as biological signals.

The GraphRAG Pipeline